

CASEY BOARD INTELLIGENCE PACK

Airport Infrastructure has been classified as Airport / Aviation in Global / Not specified. CASEY estimates \$11.5B P50 cost, \$9.2B to \$14.9B range, 85 month baseline, Medium risk and 59% confidence under the Faster scenario.

Project	Airport Infrastructure	Scenario	Faster
P50	\$11.5B	Cost Range	\$9.2B - \$14.9B
QCRA P80	\$17.9B	QSRA P80	102.19 months
Direct / Indirect / Reserve	\$11.6B / \$2.0B / \$1.0B	Risk / Confidence	Medium / 59%

Board Recommendation

- Acceleration selected: schedule compression is funded through logistics premium, parallel procurement and higher delivery intensity.
- Controls consequence: QSRA risk should be watched closely because interface density and commissioning pressure rise.
- Board decision: approve acceleration only with package-level recovery plan and explicit acceleration cost allowance.

QCRA and QSRA - Separate Tornado Drivers

Rank	QCRA Cost Driver	CBS	Mean Cost	QSRA Schedule Driver	Activity	Mean Days
1	Market escalation	01.07	\$735M	Operational phasing disruption	A1400	50.9
2	Operational phasing disruption	01.04	\$567M	Permits / approvals delay	A1800	45.4
3	Scope growth	01.06	\$541M	Scope growth	A1600	37.0
4	Design maturity gap	01.10	\$532M	Supply chain delay	A2000	33.5
5	Supply chain delay	01.11	\$463M	Productivity underperformance	A2100	33.2
6	Productivity underperformance	02.01	\$453M	Design maturity gap	A1900	32.4
7	Baggage / security systems integration	01.05	\$396M	Baggage / security systems integration	A1500	30.8
8	Interface misalignment	02.03	\$372M	Interface misalignment	P2000	23.6
9	Commissioning delay	02.02	\$303M	Commissioning delay	P1000	22.2
10	Permits / approvals delay	01.09	\$217M	Market escalation	A1700	0.0

Risk Register - Board Top Risks

ID	Risk	Cause	Event	Impact	Owner	Response
R-002	Market escalation	Supplier market tightness, inflation, FX and long lead times	Prices exceed budgeted escalation profile	P50 cost becomes understated; commercial approvals or procurement strategy may fail	Commercial Strategy	Early procurement, index-linked allowances, market testing,
R-A01	Operational phasing disruption	Works in or near live airport operations constrained by access and sequencing	Operational access and sequencing reduce productive window	Delay, night-work premiums, stakeholder disruption and ORAT freeze	ORAT Freeze	Phasing simulation, possession plan, airport ops integration
R-001	Scope growth	Ambiguous requirements, immature scope freeze approval	Approved scope expands beyond estimate freeze	Additional quantities, redesign, rework, procurement churn	Project Board	Contingency drawdown
R-004	Design maturity gap	Estimate is based on immature design, incomplete surveys	Design changes occurring in late design	Cost growth, quantity movement, rework and schedule resourcing	Design Manager	Design maturity gates, independent review, assumption register
R-006	Productivity underperformance	Access constraints, poor sequencing, labour scarcity	Productivity rates underperform plan	Extended duration, increased preliminaries and loss of float	Delivery Lead	Package productivity controls, daily planning, earned value control
R-005	Supply chain delay	Long-lead equipment, constrained suppliers or supplier promises	Equipment delivery dates move right	Delayed installation, critical path movement and acceleration costs	Procurement Lead	Alternate suppliers, early orders, expediting, framework options
R-A02	Baggage / security systems integration	Complex passenger systems integration and vendor interface	Integrated baggage systems do not pass readiness	Commissioning delay, passenger disruption and late operation	Systems Lead	Factory acceptance testing, integration lab, ORAT dry-runs
R-008	Interface misalignment	Multiple contractors, unclear interface ownership	Package interfaces do not align during installation	Rework, claims, delay and fragmented accountability	Integration Manager	Interface control documents, weekly interface board, accountability
R-007	Commissioning delay	Incomplete readiness, software integration issues	Systems failed during testing repeated commissioning cycles	Delayed handover, operational readiness slippage and liquidation	Commissioning Lead	Commissioning readiness plan, early test packs, digital system
R-003	Permits / approvals delay	Authority process, stakeholder objections or incomplete submissions	Complete submissions beyond baseline	Critical path delay, extended preliminaries and potential re-design	Design Lead	Authority plan, consent tracker, early submissions, stakeholder

Cost Estimate - Selected Class Only

CBS	Description	Type	P10	P50	P90	Basis
01.01	Site Preparation / Enabling	Direct	\$551M	\$689M	\$896M	Demolition, enabling, logistics and access
01.02	Substructure / Foundations	Direct	\$735M	\$919M	\$1.2B	Piling, groundworks, retaining and foundations
01.03	Superstructure / Civil Works	Direct	\$1.3B	\$1.6B	\$2.1B	Frame, civils, primary structures
01.04	MEP / Systems / Utilities	Direct	\$1.3B	\$1.6B	\$2.1B	Mechanical, electrical, controls and utilities
01.05	Specialist Equipment / Fit-Out	Direct	\$1.1B	\$1.4B	\$1.8B	Permanent equipment and specialist fit-out
01.06	Airfield Works	Direct	\$1.3B	\$1.6B	\$2.1B	Runway, taxiway, apron and airfield systems
01.07	Terminal / Passenger Systems	Direct	\$919M	\$1.1B	\$1.5B	Terminal, baggage, security and passenger systems
01.09	Direct Labour	Direct	\$827M	\$1.0B	\$1.3B	Productivity, supervision and trade labour
01.10	Direct Materials	Direct	\$919M	\$1.1B	\$1.5B	Concrete, steel, envelope, MEP materials
01.11	Plant / Equipment	Direct	\$368M	\$460M	\$597M	Cranes, heavy plant and site equipment
02.01	General Conditions / Prelims	Indirect	\$643M	\$804M	\$1.0B	Site management, safety, offices and temporary services
02.02	Design / Professional Fees	Indirect	\$643M	\$804M	\$1.0B	Design, QS, controls, assurance and consultants
02.03	Permits / Insurance / Bonds	Indirect	\$276M	\$345M	\$448M	Permits, inspections, legal, bonds and insurance
03.01	QCRA Contingency	Reserve	\$551M	\$689M	\$896M	Known-unknown risk allowance
03.02	Management Reserve	Reserve	\$276M	\$345M	\$448M	Sponsor-held reserve